

Fax message

**PHILIPS****Philips TV Test Equipment A/S**

To:

Ian Whithewann

Calibrations Systems

Fax: +44 1276 600 241

Copy:Fax:

From:

Rolf Søbirk

Kornmarksvej 21
DK-2605 Brøndby
Denmark

Tel: +45 32885911

Fax: +45 43432390

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Subject: PM5640, DC off-set

The expected value for the DC off-set is < 50 mV. Only in case the off-set exceeds 100 mV there is reason to suspect a fault.

There is no easy ways to correct the off set. Any improvement requires modification of the individual output amplifiers.

On the program output a large bias resistor mounted between the base of Q-411 in unit 3 and either the positive or the negative bias can provide the necessary correction.

On the other output amplifiers a similar bias connected to the inverting input of the amplifiers will do the trick.

Please find appended the technical data for PM5640.

Best regards,
Philips TV Test Equipment

Rolf Søbirk
HelpDesk Manager

3. Technical data.

3.1 Safety characteristics

This apparatus has been designed and tested in accordance with Safety Class I requirements of IEC publication 348 (Safety Requirements for Electronic Measuring Apparatus), and has been supplied in a safe condition. This manual contains information and warnings which must be followed to ensure safe operation and to retain the apparatus in a safe condition.

3.2 Performance characteristics

Properties expressed in numerical values with stated tolerances are guaranteed by the Philips organization in your country. Specified non-tolerance numerical values indicate those that could be nominally expected as a mean of a range identical instruments.

3.3 Versions.

PM 5640G/I	PAL	CCIR/UK
PM 5640M	NTSC	FCC

3.4 Operating modes

3.4.1 Var. Video Out

This output works independently of the VITS inserter, and has three modes:

- Both VITS and video signal are calibrated.
- Calibrated VITS and variable video signal.
- All signals variable.

The signal parts of the composite video signal can be controlled in amplitude independently as well as

combined. External superimposed signal can be switched in with fixed gain.

3.4.2 VITS inserter

1. By-pass mode

To protect a program from any interruption caused by a malfunction, e.g. as the result of poor or non-standard program signal, the PM 5640 has a By-pass protection circuit (By-pass relay). This By-pass relay loops the program signal directly to the program output connector in the case of one or more of the following conditions:

- The "BY-PASS" button is pressed (locally or remotely).
- Power line failure or instrument switched off.
- The program signal falls below -6dB (measured on sync amplitude) and automatic substitution is not selected.

2. Normal mode

The instrument operates in the normal mode when the program signal is $> -6\text{dB}$ (measured on sync amplitude). The program signal then passes through the instrument's inserter/deleter circuit. Internally generated VITS and signal applied to EXT DATA input (e.g. Teletext) are inserted into the program signal according to the program stored in the non-volatile memory.

If the program signal is $< -6\text{dB}$ (measured on sync amplitude) and automatic substitution is selected, the program signal is disabled and replaced by a full field test signal with burst and sync according to the program stored in the non-volatile memory.

3. Full-field mode

The program signal is disabled. The PM 5640 generates a full-field test signal with burst and sync according to the front plate selection. The available signals are listed in the PATTERN LIBRARY chapter (chapter 8).

The instrument genlocks to the EXT SYNCHR signal if present.

3.4.3 Synchronization

Depending on the operating mode the generator synchronizes to the various inputs:

- In By-pass and Normal mode the sync generator locks to the signal at PROGRAM IN, either slaving sync to sync and subcarrier to burst, or maintaining correct Sc-H phase. With 8-field reference pulse, in line 7 field 1, the generator locks into a 8-field sequence. Without the reference pulse the generator locks into a 4-field sequence.
- In Full-field mode the synchronization input is switched to EXT SYNCHR, but having the same functions as above. In this mode internal synchronization can be forced into action.

When no synchronization signal is available the generator has internal lock with correct Sc-H phase.

- In By-pass and Normal mode the phasing range of the line phase is restricted to approx. $\pm 0.5\mu\text{sec}$.

3.5 Inputs

PROGRAM IN

Impedance:

High ohmic, looped through (By-pass mode). Internally terminated with 75ohm (Normal and Full-field mode).

Amplitude:

1Vpp nominal

Acceptable input level:

Nominal $+3$ to -6dB (measured on sync amplitude)

Return loss:

$>34\text{dB}$ up to 7MHz (Normal and By-pass mode)

EXT SYNCHR

Impedance:

High ohmic, looped through

Input signals:

Composite video, Black burst, or Composite sync

Amplitude:

0.5 to 4Vpp (composite sync).

$\pm 6\text{dB}$ (composite video).

Both signals with 100% Hum (max. 1Vpp).

Return loss:

$>40\text{dB}$ up to 7MHz

EXT IN

Impedance:

High ohmic, looped through.

Gain:

1.0 $\pm 2\%$

Amplitude response:

10Hz-10MHz $\pm 2\%$

10MHz-20MHz $\pm 4\%$

Return loss:

$>34\text{dB}$ up to 15MHz

Tilt:

$<0.5\%$ (50Hz and 15kHz)

Line time non linearity:

$<0.5\%$

Differential gain:

$<0.2\%$

Differential phase:

$<0.15^\circ$

In CW mode blanking is mixed to the signal and a pedestal can be added.

EXT DATA

The input signal can have sync, but burst is not allowed.

Impedance:

High ohmic, looped through

Gain:

1.0

Amplitude response:

10Hz-5MHz (within $\pm 2\%$)

Return loss:

$>34\text{dB}$ up to 7MHz

EXT SUP IN

Impedance:

Approx. 1kohm

Gain:

Fixed, 100% of incoming signal

Maximum input amplitude:

$\pm 2\text{V}$

3.6 Outputs.

PROGRAM OUT

Impedance:

75ohm $\pm 0.5\%$ (resistive component)

Return loss:

>34dB up to 7MHz (By-pass mode)

>34dB up to 15MHz (Normal and Full-field mode)

MONITOR OUT

Impedance:

75ohm $\pm 0.5\%$ (resistive component)

Return loss:

>34dB up to 15MHz (Normal and Full-field mode)

VAR. VIDEO OUT

The specifications under this point are valid when all level controls are set to their nominal values.

Impedance:

75ohm $\pm 0.5\%$

Frequency response:

0-10MHz $\pm 1\%$

10-15MHz $\pm 2\%$

15-20MHz $\pm 5\%$

Return loss:

>34dB up to 15MHz

Differential gain:

<0.2%

Differential phase:

<0.15°

Nominal white amplitude:

700mV $\pm 1\%$

All levels are digitally defined and will be within $\pm 3\text{mV}$ of the correct value relative to the calibrated 700mV level.

Variable amplitude accuracy:

$\pm 1\%$ or 2mV, which ever is greater referred to nominal white amplitude.

Isolation between outputs:

>46dB up to 1MHz

>36dB at fsuc

Random HF noise:

<-66dB_{rms} (weighted)

Low frequency noise:

<-56dB_{pp}

50Hz square wave tilt:

<0.5%

15kHz line tilt:

<0.25%

Sync amplitude:

300mV $\pm 2\%$

Sync rise and fall time:

230nsec $\pm 12\text{nsec}$

Burst amplitude:

300mV $\pm 3\%$

Sc-H phase relative to black burst:

$\pm 10^\circ$

TEST OUT

Impedance:

75ohm $\pm 0.5\%$ (resistive component)

Return loss:

>34dB up to 15MHz (Normal and Full-field mode)

BL BURST OUT

Black signal with 8-field reference pulse (pulse removable by jumper)

Impedance:

75ohm nominal

Return loss:

>34dB up to 10MHz

Sync amplitude:

300mV $\pm 2\%$

Rise and fall time:

230nsec $\pm 12\text{nsec}$

Burst amplitude:

300mV $\pm 3\%$

SYNC OUT

Composite sync output.

Impedance:

75ohm nominal

Return loss:

30dB (typical)

Amplitude:

4.0V $\pm 0.4\text{V}$ (in 75ohm)

Rise and fall time:

200nsec

BLANK OUT

Composite blanking output.

Impedance:

75ohm nominal

Return loss:

30dB (typically)

Amplitude:

4.0V $\pm 0.4\text{V}$ (in 75ohm)

Rise and fall time:

200nsec

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BK OUT

Burst key signal output.

Impedance:

75ohm nominal

Return loss:

30dB (typical)

Amplitude:

4.0V +/-0.4V (in 75ohm)

Rise and fall time:

200nsec

Fv OUT

Sync generator output (field pulses only).

Impedance:

75ohm nominal

Return loss:

30dB (typical)

Amplitude:

4.0V +/-0.4V (in 75ohm)

Rise and fall time:

200nsec

Fh OUT

Sync generator output (line pulses only).

Impedance:

75ohm nominal

Return loss:

30dB (typical)

Amplitude:

4.0V +/-0.4V (in 75ohm)

Rise and fall time:

200nsec

ID OUT

Providing color identification or PAL identification pulses

(Jumper selection between PAL-ID or 8-field pulse).

Impedance:

75ohm nominal

Return loss:

30dB (typical)

Amplitude:

4.0V +/-0.4V (in 75ohm)

Rise and fall time:

200nsec

SUBC

Continuous color subcarrier output.

Impedance:

75ohm nominal

Return loss:

>30dB at fsubc

Amplitude:

>1.0V (in 75ohm)

3.7 Remote.**3.7.1 IEEE-488 interfacebus,
PM 9191 module**

All instrument functions can be remotely controlled over the bus.

The interface complies with the specifications of the IEC-625-1 standard, except for the theoretical demand for 100usec reaction time on REN (remote enable) false. The interface is fitted with a connector complying with IEEE-488.

Interface functions implemented:

- Source handshake (SH1)
- Acceptor handshake (AH1)
- Talker function (T6)
- Listener function (L3)
- Service request (SR1)
- Remote/local (RL1)

Used Code:

ISO 7-bit code (ISO 646)

Input separator:

Fully programmable. Default separator at power on: NL (LF) and the END message (EOI). One of them, or NL and END combined, is accepted.

Output separator:

Equal to input separator, except that END (EOI) is always sent.

Connector:

24 pin female connector with contact assignment in accordance with IEEE-488.

Timing:

Communication speed is max. 1 kbyte/sec.

3.7.2 RS-232 interface with PM 9190 module (Option)

3.8 Signal generator

All signal components (Pulses, Transitions and sinewaves) are digitally generated. This assures precise timing and amplitude accuracy. However, this digital principle makes timing and amplitude to discrete functions with a certain resolution.

Clock frequency:

- 27MHz (for luminance generator)
- 7.75MHz (for chrominance generator)
- 54MHz (for zoneplate generator)

Amplitude resolution:

- 10bit (for luminance and chrominance generator)
- 8bit (for zoneplate generator)

The test signals are grouped in 5 classes:

- 100-199 Square wave signals
- 200-299 Pulse & Bar signals
- 300-399 Sine Wave signals
- 400-499 Linearity signals
- 500-599 Color + black/white signals

The test signals 100-599 are placed in a three dimensional matrix of 5 groups by 10 columns by 10 in depth.

A summarized listing of the test pattern located in the instrument is described in the PATTERN LIBRARY (chapter 8).

User defined signal:

1-9 or with extended RAM 1-25.

3.9 Signal specifications

3.9.1 Square Wave signals

0.25Hz, 50Hz, 15kHz, 250kHz (2T shape):

Amplitude:

700mV +/-1%

Rise and fall time:

200nsec +/-20nsec

Overshoot:

<0.5%

Tilt:

<0.5%

15kHz, 250kHz (T shape):

Amplitude:

700mV +/-1%

Rise time:

100nsec +/-10nsec

Overshoot:

<1%

Tilt:

<0.5%

15kHz, 250kHz (T/2 shape):

Rise time:

50nsec +/-10nsec

3.9.2 Various pulse & bar signals

2T pulse:

Amplitude:

within 1% of white bar

Width:

200nsec +/-6nsec

K factor:

<0.25%

T pulse:

Amplitude:

700mV +/-1%

Width:

100nsec +/-4nsec

K factor:

<0.5%

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10T pulse:

Amplitude:
within 1% of white bar
Width:
1000nsec $\pm$ 40nsec
Bottom curvature:
<1.5%

Chroma 20T + bar:

Amplitude:
within 1% of white bar
Chroma rise time:
330nsec $\pm$ 30nsec
Pedestal rise time:
200nsec $\pm$ 10nsec
Chroma phase:
0°

20T composite pulse:

Amplitude:
within 1% of white bar
Width:
2000nsec $\pm$ 60nsec
Bottom curvature:
<1%
Modulation unbalance:
<3mVpp
Subcarrier distortion:
<1%

Staircase:

Number of levels:
6 (black and white included)
Amplitude:
within 1% of white bar
Rise time:
235nsec $\pm$ 15nsec
Step inequality:
<0.5%

Three level chrominance bar:

Amplitude:
20%, 60%, and 100% of 0.70VPP $\pm$ 1%
Grey level:
350mV $\pm$ 1%
Rise and fall time:
1usec $\pm$ 5%
DC content:
<0.5%

One level chrominance bar, Pattern 211:

Amplitude:
within 1% of white bar
Grey level:
350mV $\pm$ 1%
Rise and fall time:
1usec $\pm$ 5%
DC content:
<0.5%

Chrominance reference:

Amplitude:
60% of 0.70VPP $\pm$ 1%
Grey level:
350mV $\pm$ 1%
Rise and fall time:
1usec $\pm$ 5%

Eye test signals:

UK 3-test signal
Chroma bar amplitude:
700mV
Chroma bar phase:
60°, PAL shifted Teletext signal
Amplitude:
462mV $\pm$ 1%

3.9.3 Sine wave signals

YITS line 18:

WHITE REFERENCE BAR:
Amplitude:
80% of 0.70VPP $\pm$ 1%
Rise and fall time:
200nsec $\pm$ 10nsec

BLACK REFERENCE BAR:

Amplitude:
20% of 0.70VPP $\pm$ 1%
Rise and fall time:
200nsec $\pm$ 10nsec

GREY LEVEL:

Amplitude:
350mV $\pm$ 1%

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Frequencies:

0.5MHz, 1.0MHz, 2.0MHz, 4.0MHz, 4.8MHz, 5.8MHz

Freq. accuracy:

+/-1%

Start/Stop phase:

0°

DC content:

<0.5%

Burst envelope rise time:

200nsec +/-20nsec

Field sweep:**Frequency span:**

-0.1MHz to 11MHz (pattern 310)

-0.1MHz to 20MHz (pattern 313)

Markers:

-1MHz for 10MHz sweep

-2MHz for 20MHz sweep

Reference bar amplitude:

700mV +/-1%

Frequency response:

-0.1MHz to 10MHz +/-1%

-10MHz to 15MHz +/-2%

-15MHz to 20MHz +/-5%

Line sweep:**Frequency span:**

-0.1MHz to 6MHz

-0.1MHz to 11.2MHz

-0.1MHz to 18MHz

Reference bar amplitude:

700mV +/-1%

Frequency response:

-0.1MHz to 10MHz +/-1%

-10MHz to 15MHz +/-2%

-15MHz to 20MHz +/-5%

Fixed frequencies:**Frequency:**

1MHz to 10MHz in steps of 1MHz

Frequency accuracy:

+/-1%

Reference bar amplitude:

700mV +/-1%

Variable frequencies:**Frequency range:**

12kHz to 20MHz in steps of 6kHz

Frequency accuracy:

+/-6kHz

Reference bar amplitude:

700mV +/-1%

Frequency response:

-0.1MHz to 10MHz +/-1%

-10MHz to 15MHz +/-2%

-15MHz to 20MHz +/-5%

Multipulse signals:**Frequencies:**

1MHz, 2MHz, 3MHz, 4MHz, 5MHz

Amplitude:

within 1% of white bar

SineX/X signal:**Bandwidth:**

0 to 5MHz +/-2%, positive and negative pulses

Zoneplate signals:**Type:**

Circular zone plate signal

Range:

-Up to 6.5MHz (pattern 370)

-Up to 13.5MHz (pattern 372)

3.9.4 Linearity signals**UK VITS signals****10T COMPOSITE PULSE:****Amplitude:**

within 1% of white bar

Width:

1000nsec +/-40nsec

Bottom curvature:

<1.5%

Modulation unbalance:

<3mVpp

Subcarrier distortion:

<1%

SUPERIMPOSED STAIRCASE**Chroma amplitude:**

140mV +/-3%

Chroma phase:

60° PAL-shifted

Differential gain:

<0.2%

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Differential phase:

<0.2°

Sawtooth + Chroma:**Luminance amplitude:**

700mV +/-1%

Chroma amplitude:

280mV +/-2%

Chroma phase:

0°

Special Sawtooth:**Luminance amplitude:**

-150mV to 700mV +/-1%

Chroma sawtooth:**Pedestal amplitude:**

350mV +/-1%

Chroma amplitude:

0 to 700mV

Chroma phase:

0°

VITS line330**STAIRCASE:****Number of levels:**

6 (black and white included)

Amplitude:

Within 1% of white bar

Rise time:

235nsec +/-15nsec

Step inequality:

<0.5%

SUPERIMPOSED SUBCARRIER:**Amplitude:**

0.28Vpp +/-2%

Rise and fall time:

1usec +/-5%

Phase:

60° +/-5° for burst 135°/225°

COMPOSITE STAIRCASE:**Differential gain:**

<0.2%

Differential phase:

<0.2°

5 step staircase + chroma:**Number of levels:**

6 (black and white included)

Amplitude:

700mV +/-1%

Step inequality:

<0.5%

Chroma amplitude:

280mV +/-2%

Chroma phase:

0°

10 step staircase + chroma:**Number of levels:**

11 (black and white included)

Amplitude:

700mV +/-1%

Chroma amplitude:

280mV +/-2%

Chroma phase:

0°

Chroma staircase:**Pedestal:**

350mV +/-1%

Number of levels:

8

Chroma amplitude:

0 to 700mV

Chroma phase:

0°

CISPR staircase:**Number of levels:**

10 (black and white included)

Amplitude:

700mV +/-1%

Staircase special:**Number of levels:**

6 (ultra black and white included)

Amplitude:

-100mV to 700mV

Chroma amplitude:

280mV +/-2%

Chroma phase:

0°

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3.9.5 Color signals

Color bar:**Number of bars:**

8

Sequence:White, Yellow, Cyan, Green, Magenta, Red, Blue,
Black**Luminance amplitude:**700mV $\pm$ 1%**Chroma amplitude:**According to standard $\pm$ 2%**Split color bar:****Number of bars:**

Identical to color bar

Split field:

2:1

Red:**Luminance amplitude:**157mV $\pm$ 3mV**Chroma amplitude:**664mVpp $\pm$ 2%**Flat-field signal:****Amplitude:**700mV $\pm$ 1%**Cross-hatch:****Pattern:**

14 x 19 lines

Amplitude:700mV $\pm$ 1%**Circle signal:**

79% of picture height with corner markings

3.9.6 Various Black and White signals

Reference signal:**Luminance amplitude:**700mV $\pm$ 1%**Pedestal amplitude:**350mV $\pm$ 1%**Chroma amplitude:**700mV $\pm$ 2%**Chroma phase:**

180°

PLUGE signal:**Ultra black amplitude:**

-14mV

Dark grey amplitude:

14mV

Black level:

0mV

Grey scale levels:

700mV, 441mV, 210mV, 112mV

3.9.7 Antipal signal

Pedestal amplitude:

442mV

Chroma amplitude:

377mVpp (G-Y field)

212mVpp (R-Y field)

312mVpp (B-Y field)

Chroma phase:

G-Y for yellow reference

3.9.8 Testpattern

Vertical multiburst 0.5MHz, 1.3MHz, 2.3MHz,
4.2MHz, 4.8MHz, 5.8MHz (with 100% white bar re-
ference)Pluge signal $\pm$ 2%

2T + 20T composite signal

EBU colorbar signal

Staircase signal

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3.10 Amplitude controls (Var. Video Out)

The signal components of the composite signal can be controlled in amplitude via the frontplate controls or the remote interface.

3.10.1 Video

Amplitude range:

0 to 1000mVpp

Nominal level:

700mVpp

The video signal can be switched off.

3.10.2 Set-up

Amplitude range:

-300mVpp to 900mVpp

The Set-up signal can be switched off.

3.10.3 Sync

Amplitude range:

0 to 600mVpp

Nominal level:

300mVpp

The Sync signal can be switched off.

3.10.4 Burst

Amplitude range:

0 to 600mVpp

Nominal level:

300mVpp

The Burst can be switched off.

3.10.5 Composite video

Amplitude range:

+3dB to -60dB

Nominal level:

0dB

The composite video signal can be switched off or inverted.

3.10.6 Average Picture Level (APL) in duty mode

Amplitude range:

0 to 700mV

Sequence:

1+3 (one line with the selected signal followed by three lines containing the APL level)

3.10.7 Noise (white)

Noise spectrum (bandwidth limiter):

Flat from 10kHz to 5MHz

Amplitude control:

-10 to -60dB_{RMS} (calibrated in 5MHz BW)

Accuracy of setting:

+/-2dB

3.10.8 Hum

Internally generated, 50Hz mains derived

Amplitude range:

0 to 1.5Vpp

3.10.9 Signal in vertical interval

Test signals as programmed (see operating part of this manual) with calibrated amplitudes.

Second function sets VITS in variable amplitude mode.

3.11 VITS insertions

3.11.1 Insertion

Range:

- Line 6 and 8 to 22 (included). If PAL-ID pulse is switched off, then line 7 may also carry VITS information.
- Line 319 to 335 (included)
- Line 7 carries 8 field reference (if selected)
- All information in lines to be inserted upon is automatically erased before insertion takes place.

Available signals:

The flexibility has been enhanced in two ways:

- Selection between 7 factory installed VITS groups, group 1 and 2 are the previous G and I-version.

For the complete range, please refer to Chapter 6, Installation.

- Selection between all lines in line library 1-255 is possible in a user-defined VITS-group. Line type 250 to 255 are user definable lines.

For the complete range, please refer to Chapter 9, Line library. For programming instructions, please refer to Chapter 7, Operating instructions.

Restrictions: Extensive use of chroma RAM for VITS-lines might exhaust the memory available for the full-field signal, however for normally used signals it will not occur.

Note: Line 7 field 1 which normally carries the 8-field reference is indicated like other lines in the display when programming VITS lines but can only be use as a VITS line when the 8-field reference is disabled.

3.11.2 Erasure

Range:

- Line 6, 7 (if PAL-ID pulse is switched off) and 8 to 22 (included)
- Line 319 to 335 (included)
- Line 7 carries 8 field reference pulse (if selected)

3.11.3 Characteristic in "Normal" mode

Isolation between transmission and AUX outputs:

>46dB up to 1MHz

>36dB at fsubc

Gain:

1.00 +/-1%

FREQUENCY RESPONSE:

"Normal" mode:

-Up to 10MHz +/-1%

-10MHz to 15MHz +2/-5%

-15MHz to 20MHz typ. -5%

"Program out" in by-pass mode:

-Up to 6MHz +/-1%

-6MHz to 10MHz +1/-5%

Chrominance/Luminance inequalities:

<0.5% (gain)

<+/-5nsec (delay)

2T pulse to bar ratio:

100% +/-0.25%

2T pulse overshoot:

Must not differ from the input signal by more than 0.5% of the luminance bar amplitude

50Hz square/wave tilt:

<0.5%

15kHz line tilt:

<0.25%

Differential phase:

<0.15 (at standard level)

<0.3% (at +3dB)

Differential gain:

<0.2% (at standard level)

<0.4% (at +3dB)

Line time non linearity:

<0.25%

Random HF noise:

<-75dB_{RMS} (weighted)

Residual subcarrier on program line:

<-60dB_{pp}

Hum and lower order harmonics:

<-60dB_{pp}

Spurious transients during active lines:

<-60dB_{pp} (up to 10MHz)

Spurious transients outside active lines:

<-40dB_{pp} (up to 10MHz)

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SIGNAL ATTENUATION IN ERASE MODE:**2T pulse:**

>70dBpp

Subcarrier:

>60dBpp

CROSS TALK INTO MAIN SIGNAL:**2T pulse:**

<-70dBpp

Subcarrier:

<-60dBpp

Test line video amplitude:

0.7Vpp

**Shift of signal delay through instrument by
change from Normal to By-pass mode:**

<1.5nsec

**3.11.4 Characteristic in Full-field and
substitution mode****White level amplitude:**

700mV +/-1%

Frequency response:

-Up to 10MHz +/-1%

-10MHz to 15MHz +/-5%

-15MHz to 20MHz typ. -5%

Random HF noise:<70dB_{RMS} (weighted)**Residual subcarrier:**

<2mVpp

Hum and lower order harmonics:

<-60dBpp

Spurious transients during active lines:

<-60dBpp (up to 10MHz)

Spurious transients outside active lines:

<-40dBpp (up to 10MHz)

Sc-H phase relative to black burst:

+/-10°

3.12 Sync pulse generator**3.12.1 Internal mode**

The sync pulse generator is controlled by internal
X-tal oscillator.

COLOR SUBCARRIER:**Frequency:**

4.43361875MHz

Adjustment tolerance:

+/-0.5Hz

Temperature stability:<1x10⁻⁶ (0-50°, ref. 25°C)<3x10⁻⁷ typical (25-35°C)**Ageing:**<1x10⁻⁷ per month**LINE/SUBCARRIER PHASE****(BLACK BURST OUTPUT):****Absolute phase:**

+/-5° at 25°C

Jitter:

+/-2°

Drift:

+/-2°/10°C

COMPOSITE SYNC:**Line sync width:**

4.7µsec +/-0.2µsec

Front porch width:

1.5µsec +/-0.2µsec

Equalizing pulse:

2.35µsec +/-0.1µsec

Serration pulse width:

4.7µsec +/-0.2µsec

Number of serration pulses:

5

Number of equalizing pulses:

5 + 5

COMPOSITE BLANKING:**Line blanking duration:**

12.0µsec +/-0.3µsec

Field blanking duration:

25lines +12µsec

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BURST:**Width:**2.25µsec $\pm$ 0.23µsec**Position:**5.6µsec $\pm$ 0.1µsec (related to leading edge of line sync)**Burst suppression for field 1 to 4:**

- Line 623 to 6 inclusive (field 1)
- Line 310 to 318 inclusive (field 2)
- Line 622 to 5 inclusive (field 3)
- Line 311 to 319 inclusive (field 4)

Jitter:

<1° (with respect to incoming Burst phase)

Subcarrier phase range:

>360° (via front panel)

Stability of phase:Typically $\pm$ 2°**3.12.2 External mode**

The sync generator genlocks to an external video or composite sync signal.

Three modes are available:

- a. Normal slaving, i.e. sync to sync and subcarrier to burst
- b. Constant Sc-H phase mode in a 8-field sequence (requires that genlock signal carries a 8-field reference pulse in line 7 field 1)
- c. Constant Sc-H phase mode in a 4-field sequence (no 8-field reference on the genlock signal)

Sync-lock:

Crash lock in By-pass and Normal mode. Slow lock in Full-field mode (locking to external source).

SYNC-LOCK DATA FOR SLOW LOCK:**Horizontal frequency lock range:** $\pm$ 10ppm**Lock-in time (vertical):**

<7sec

Jitter (with respect to input sync):

<6nsec (for noise free signal with nominal frequency and amplitude)

Line phase change:<15nsec (for sync level 300mV $\pm$ 6dB)**Line phase adjustment:** $\pm$ 12/-3µsec (via front panel)**SUBCARRIER LOCKING:****Lock range:** $\pm$ 25Hz**Lock-in time:**

<1sec

3.13 Survey of programmable functions**3.13.1 Generator control commands**

Pattern select

Luminance on/off

U on/off

V on/off

Scroll on/off

Duty on/off

Jump on/off

Jump rate select

Color ID on/off

Field off

Cross-hatch position

Recall/Select

Store

VITS function:

By-pass

Normal

Full-field

Full-field synchronization

Substitution pattern

25Hz PAL offset on/off

Crash lock on/off

Constant Sc-H phase on/off

External color ID lock on/off

External carrier lock

System select

3.13.2 Generator programming commands

Pattern definition
User defined zoneplate
User defined sweep
User marker select
User reference select
User line definition
VITS reset
VITS programming
VITS line status

3.13.3 Variable video output control commands

Standard
Display mV/dB
Video level
Sync level
Burst level
Set-up level
Composite level
APL level
Noise level
Hum level
Frequency
Inverted video on/off
Sync phase
Subcarrier phase
Sc-H phase
Noise gate on/off
Superimposition

3.13.4 Output (request) data

Measuring data:

VITS status
Frequency
Jump rate
Amplitude mV
Amplitude dB
Integer data
Status on/off

Identity:

Device status data

3.13.5 Miscellaneous functions

2nd XX SELECT CODES:

- 10: Color ID pulse on (default)
- 11: Color ID pulse off
- 12: Scroll on
- 13: Scroll off (default)
- 14: PAL offset on (default)
- 15: PAL offset off
- 16: Crash lock on (default in Normal and By-pass mode)
- 17: Crash lock off (default in Full-field mode)
- 18: External carrier lock on (default)
- 19: External carrier lock off
- 20: Constant Sc-H phase on (default)
- 21: Constant Sc-H phase off
- 22: External color ID lock on (default)
- 23: External color ID lock off
- 24: Nominal VITS amplitude on "Var video out" (default)
- 25: Variable VITS amplitude on "Var video out"
- 26: VITS is switched on in "Full-field" mode
Switching to "Normal" or "By-pass" restores VITS
- 27: VITS is switched off in "Full-field" mode
- 30: Duty signal locked to field frequency
- 31: Duty signal unlocked.
Note: VITS group selection reset to default (locked for I-version)
- 87: Error monitor on/off (off is default)
- 90: Standard VITS set-up

The next codes 91, 92, and 93 are codes for storing factory adjusted phases, as these cannot be overwritten accidentally anymore.
Codes 92 and 93 are used only when a complete calibration is needed and can only be activated once and only after a "Factory reset".

- 91: Store the adjusted the H(orizontal) and S(ubc) phase of the inserted VITS signal in the "Normal" and "By-pass" mode.
- 92: Store gen-lock phase adjustment.
- 93: Store internal phase adjustment.

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3.14 Environmental conditions

The environmental data mentioned in this manual is based on the results of the manufactures's procedure.

Details of these procedures and failure criteria are supplied on request by the PHILIPS organization in your country, or by PHILIPS, INDUSTRIAL ELECTRONICS DIVISION, EINDHOVEN, THE NETHERLANDS.

3.14.1 Climatic conditions

Operating temperature:

+5°C to +45°C

Storage temperature:

-30°C to +70°C

Climatic class:

C1

3.14.2 Mechanical requirement

VIBRATION

Limit range for storage and transport:

30min. in each of three directions, 10Hz to 150Hz;

0.7mm/p and 50m/s² max. acceleration.

According to IEC publication 68, test Fc

NOTE: Unit mounted on vibration table without shock absorbing material.

BUMP

Limit range for storage and transport:

1000 bumps of 100m/s² sine, 6ms duration in each of three directions.

According to IEC publication 68, test Eb

PACKING

According to UN-D-1400

The test methods mentioned in the N.V. Philips standard UN-D-1400 are in accordance with those of relevant ISO standards.

3.15 Mains supply conditions

Mains supply voltage:

100V, 120V, 220V, 240V +/-10% (max. 250V)

Mains frequency:

48Hz to 65Hz

Power consumption:

65W (at 220V)

3.16 Mechanical dimensions

Cabinet type:

2U high, 19 inch rackmount/table cabinet

Maximum dimensions:

Height : 88mm (3.5 inch)

Depth : 421mm (16.6 inch) - in 19 inch rack

Depth : 466mm (18.3 inch) - incl. front brackets

Width : 440mm (17.3 inch) - in 19 inch rack

Width : 483mm (19.0 inch) - incl. front brackets

Weight : 8kg (18lbs)